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RESEARCH-ARTICLE

Automating open educational resources assessments: a machine learning generalization study

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Automating Open Educational Resources Assessments: A Machine Learning Generalization Study

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ABSTRACT

Assessing the quality of online educational resources in a cost effective manner is a critical issue for educational digital libraries. This study reports on the approach for extending the Open Educational Resource Assessments (OPERA) algorithm from assessing vetted to peer-produced content. This article reports details of changes to the algorithm, comparisons between human raters and the algorithm, and the extent the algorithm can automate the review process.

Categories and Subject Descriptors

K.3 [Computers and Education]: Computer Uses in Education - Collaborative Learning

General Terms

Design, Reliability, Human Factors, Standardization.

Keywords

Education Digital Library, Resource Quality, Machine Learning.

1. INTRODUCTION

The increased pervasiveness of networked computing coupled with a participatory web culture has spawned new models of content creation. These models, variously referred to as collective intelligence, crowd sourcing, or peer production, occasionally take the Internet by storm [2]. As canonical examples, *Wikipedia* and *YouTube* need no introduction.

Similarly, in education, the scalable deployment of media-rich online resources supports peer production in ways that promise to radically transform teaching and learning [4, 10]. For example, a growing trend toward sharing online instructional resources has spawned the global *OpenCourseware* movement [13]. Likewise, digital libraries such as the National Science Digital Library (NSDL.org) and the Digital Library for Earth Systems Education (DLESE.org) collect and curate online learning resources created for a wide range of educational audiences and subject areas [8].

We have developed a web-based authoring tool, called the Instructional Architect (IA.usu.edu), which enables teachers to create instructional activities for their students using online learning resources from the NSDL and the wider Web [11].

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Teachers can share these resulting activities, called *IA projects*, by making them available for just their students or publically on the Web. These IA projects can then be viewed, copied, and adapted by other IA users, in ways that support innovative teacher peer production. Currently, the IA has over 5,700 registered users who have created over 12,600 IA projects using over 57,300 online resources.

A vexing problem for peer production environments remains the elusive notion of *quality*. In these environments, how does one identify quality online content? Moreover, how does one do so in sustainable, cost-effective, and scalable ways? Previous work [1] presented an innovative approach for using machine learning models to automatically assess the quality and pedagogic utility of educational digital library resources. The resulting set of models, called OPERA, demonstrated the feasibility and accuracy of automatic quality assessments for a single STEM domain and audience-level: high school Earth science. In this article, we report on recent efforts to extend these models to support a broader range of STEM topics and grade levels. Specifically, we applied the quality models to 200 IA projects and compared model outputs to quality assessments made by K-12 teachers. Since the nature of the resources being compared in the IA (peer) versus DLESE (expert) are different, results of this study provide insights on the generalizability of this machine learning approach and its potential for facilitating teacher peer production.

2. PRIOR WORK

2.1 Quality in Educational Digital Libraries

Many digital libraries have established review rubrics to evaluate the quality of online educational resources generated by users, including teachers and faculty [e.g., 5, 7]. Some libraries have developed rubrics to identify high quality online resources in order to establish their reputation with users [14], while others use quality rubrics to determine the inclusion of resources in a digital library or repository [7]. One site developed a rubric to help guide authors in creating high quality online resources [9] by gathering feedback from target users. As these examples indicate, assessing the quality of submitted resources remains a critical issue for educational digital libraries. All of these approaches rely heavily on significant human effort to assess resources, and as such, are difficult and expensive to implement at scale.

2.2 Quality Machine Learning Models

The Open Educational Resource Assessments (OPERA) used in this study were initially developed for assessing quality in online resources collected in the Digital Library for Earth Systems Education [1]. Acknowledging that quality is contextual, this approach avoided a unary “thumbs-up/thumbs-down” assessment;

instead it relied on characterizing quality as decomposable into a number of indicators, with can be applied individually or in combination. These indicators were identified and distilled using a detailed expert study protocol involving science educators and science librarians (see Table 1).

Table 1. Quality indicators [1]

Quality Indicators	Baseline Performance	ML Model Performance
Not inappropriate for age	99%	99%
Indicates age range	79%	87%
Has instructions	61%	78%
Identifies learning goals	72%	81%
Organized for goals	75%	83%
Has prestigious sponsor	70%	81%

Each indicator is modeled using a support vector machine, a type of supervised machine learning algorithm. Supervised machine learning algorithms construct models by statistically analyzing a training corpus for which the correct judgment is known. The OPERA models were trained on a set of human-tagged DLESE resources that indicated the presence or absence of each indicator. Through training, the models attempt to learn which features of an online resource help to determine the presence or absence of an indicator. While the support vector machine algorithm has been shown to be effective at detecting relevant statistical patterns even when the number of features is extremely large, the way in which those features are presented to the algorithm greatly influences the resulting model. The initial set of features used in this study was guided by a large corpus of prior work in using machine learning on linguistic and semantic tasks: features used include ‘bag-of-words’, term frequency, resource URLs, Google page rank, etc.

Evaluation took place using a different set of DLESE resources from the training corpus. Each model’s output was compared to the ratings of two expert DLESE catalogers who were asked to judge the presence or absence of the quality indicators in each resource in the evaluation set. Table 1 shows six of the quality indicators used and the percent of time the models agreed with the human experts. Model results are also compared to a simple baseline that always assumes the most common case. For example, the “has instructions” indicator is present in 39% of resources. If we always assume that a resource has no instructions, we’d be correct in 61% of cases. Good improvements over the baseline were achieved on the “has instructions” and “has prestigious sponsor” indicators, and moderate improvements on the “indicates age range” and “organized for goals” indicators.

3. RESEARCH DESIGN AND METHODS

The purpose of this study was to formulate an approach for identifying high quality IA projects, using the quality indicators identified by [1], and operationalized in the OPERA machine learning models. Three teachers rated IA projects using the quality indicators and their ratings were compared to OPERA output. The following questions guided this study:

1. How do teacher IA project ratings compare with each other?
2. How do teacher IA project ratings compare with OPERA?
3. To what extent can the OPERA algorithm be used to identify ‘quality’ IA projects?

3.1 Participants and Procedures

Three experienced science teachers who have used the IA and participated in a teacher professional development workshop participated in the study. They were asked to individually rate 200 randomly selected IA projects. This was done through the use of a side bar add-on in Mozilla Firefox, in which they were presented with the IA project on the right side of the screen and the six indicators on the left. The teachers clicked radio buttons under the indicators to select their choice (1-5 scale from “strongly disagree” to “strongly agree”) for each indicator.

The 200 IA projects selected for rating 1) were publicly available, 2) were viewed at least 20 times, 3) had more than 700 words, 4) used at least 3 online resources, and 5) were related to science education. In this way, the selected IA projects represented a multiple science topics at multiple grade levels, and of varying quality. Six quality indicators were selected (see Table 2) from the initial seven reported in [1].

Table 2. Six indicators for rating IA projects

Quality Indicator	Definition
Has instructions	Tells user how to navigate/use the project
Links to prestigious sponsor	Links to 'prestigious' source or site with highly respected organizer
Identifies learning goals	Identifies learning goals, articulates the knowledge/skills a student will acquire
Organized for learning goals	Organizes content appropriately for its learning goals
Identifies age range	Identifies its target student age range by stating the expected age or grade
Content appropriate for age range	Provides reading/activities that are neither too difficult nor easy for the given range

3.2 OPERA Algorithm

The algorithm previously developed [1] was used to classify the same 200 IA projects along 6 quality indicators. Two versions of the OPERA algorithm were used. First, ‘out-of-the-box’ OPERA was the same as used in previous research [1]. Second, trained OPERA was a version trained on ½ of the IA projects (100 projects), and then asked to classify the remaining 100 projects.

OPERA provides two outputs for each quality indicator on each IA project: a yes/no classification as to whether the IA project possesses the indicator, and a score ranging from 0 to 1, indicating the confidence OPERA has in the yes/no response. This score can be viewed as a pseudo-probability, in that the closer to 0 the score is, the more confidence that the answer is "no", and the closer to 1, the more confidence that the answer is "yes".

3.3 Measuring Agreement

Machine learning research has typically relied on the *kappa* statistic to measure agreement, and is robust for nominal data [3]. However, *intraclass correlation* (ICC) and *Krippendorff’s alpha* (KA) are both more appropriate for interval data [6, 12]. An ICC statistic reports single and average measures, where the single measure takes into account the object being rated while the average measure accounts for both the object and the difference between raters. The measure in this study used an interval scale (from one to five) applied by three teachers. To account for rater and object variability, we therefore chose to use the ICC statistic

4. RESULTS

4.1 How do teacher IA project ratings compare with each other?

Table 3 shows the distribution of ratings for each teacher. They suggest a possible ceiling effect as each of the teacher's ratings have a negative skew.

Table 3. Descriptive statistics for each teacher rater (1= "Strongly Disagree"; 5= "Strongly Agree").

	Minimum	Maximum	Mean	SD
Teacher 1	1	5	4.05	1.53
Teacher 2	1	5	4.06	1.14
Teacher 3	1	5	4.07	1.24

To measure teacher agreement, average and single intra-class correlations (ICC) between the three teachers' ratings were computed for each indicator (see Table 4; 0.40 to 0.59 represents moderate inter-rater reliability, 0.60 to 0.79 substantial, and 0.80+ outstanding). The overall average ICC was .622, a substantial level of inter-rater reliability. All average ICC values achieved at least *moderate* levels of agreement, except for "links to prestigious sponsor". Two indicators, "content seems appropriate for age range" and "identifies age range", achieved *substantial* levels of agreement. Interestingly, the rank order of the teachers' average ICC on indicators is identical to the agreement between experts used in the previous study. Overall, single ICC values were substantially lower than average ICC values, suggesting that reliability performance improves with multiple raters.

Table 4. ICC value for teacher ratings on each indicator

Quality Indicator	Average ICC for teachers' ratings	Single ICC for teachers' ratings
Content appropriate for age range	.717	.458
Identifies age range	.677	.411
Has instructions	.566	.303
Identifies learning goals	.512	.259
Organized for learning goals	.433	.203
Links to prestigious sponsor	.297	.123
<i>Overall</i>	.622	

4.2 How do teacher IA project ratings compare with OPERA?

Table 5 shows, for each indicator, the % of IA projects where *all* three teachers gave a '5' rating, the % of IA projects identified as exhibiting the indicator by the 'out-of-the-box' OPERA and by the trained OPERA, and the overall correlation between the median of the teachers' ratings and trained OPERA's probability score. The indicators are ranked by teacher ICC values (see Table 4). For example, for the "content seems appropriate for age range" indicator, teachers showed high agreement between their ratings (.717); teachers' ratings identified 81.5% of the IA projects as possessing that indicator; the 'out-of-the-box' OPERA identified 95% of IA projects as possessing that indicator; and trained OPERA identified 85% as exhibiting that indicator; finally, the

correlation between median teacher ratings and the probability score is moderate at .56.

Table 5. Teacher and OPERA classifications, and correlations

Quality Indicator	Has indicator: teachers	Has indicator: 'out-of-box' OPERA	Has indicator: trained OPERA	<i>r</i> (teacher rating and trained OPERA)
Content appropriate for age range	81.5%	95%	85%	0.56
Identifies age range	94%	3%	79%	0.49
Has instructions	91.5%	84%	85%	0.44
Identifies learning goals	79.5%	3%	44%	0.44
Organized for learning goals	84%	95%	95%	0.72
Links to prestigious sponsor	53.5%	0%	5%	0.12

From Table 5, we note that with training, OPERA's identification of indicators increases, and better matches those of the teachers. Trained OPERA shows good similarities in classification on two indicators: "content seems appropriate for age range" and "has instructions". Correlations between human and machine ratings are also good for those two indicators. Both versions of OPERA appear to show little discrimination power on two indicators: "organized for learning goals" and "links to prestigious sponsors". Recall that these two indicators also showed the lowest ICC values between the teachers, and it may be that OPERA has no strong model upon which to base its decision.

A multiple regression was run to determine the degree to which the OPERA's confidence scores reflected teacher ratings (1-5), and the influence of the six different categories. The independent variable was the median teacher rating, while dependent variables included OPERA confidence score (0-1), and the rating category. A total of 660 ratings (110 for each category) were used in the model. The final regression ($R^2 = .33$, $F(3, 657) = 108.09$, $p < .01$) suggests three variables are predictive of teacher ratings at a statistically significant level. OPERA score ($\beta = 2.15$, $p < .01$) is positively related to teacher ratings, and prediction improves when ratings are for either "content seems appropriate for age range" ($\beta = 0.92$, $p < .01$) or "has instructions" ($\beta = 0.27$, $p < 0.02$). In terms of practical significance, about 33% of the variability in teacher ratings is accounted for by these two ratings categories, and the OPERA score. Given the variability in teacher ratings themselves, this is a substantial amount of predictive power.

4.3 To what extent can OPERA be used to select 'quality' IA projects?

As noted, trained OPERA's output better matched the teachers than 'out-of-the-box' OPERA, especially on two indicators ("content

seems appropriate for age range” and “has instructions”). These were also indicators with good teacher agreement. Thus, trained OPERA could perhaps be used to automatically detect quality along these two indicators.

OPERA’s performance was poor on “links to prestigious sponsor”, but this indicator also had the lowest teacher ICC value, suggesting a lack of agreement among teacher raters. Thus, is an indicator with little agreement among teachers a reasonable measure of quality? We believe it may simply be too subjective, and should not be left to an algorithm to determine.

5. CONCLUSION

This article has addressed the thorny and complex problem of measuring quality in peer produced online products. Following previous research, we chose to not define quality as a binary “thumbs-up/thumbs-down” construct; rather we acknowledge quality is defined by a confluence of sometimes subjective indicators.

Not surprisingly, study results showed that human raters sometimes agree and disagree about quality once decomposed into different indicators. Results also showed that this was a hard task for teachers, and that reliability results improve with multiple raters.

We also investigated extensions to and training of a machine learning algorithm, OPERA. This algorithm was previously developed to automatically assess quality in vetted resources in an educational digital library specifically targeted at high school earth science. In particular, we investigated the extent that OPERA generalizes to a wider range of resources, thereby serving as a proxy for the expensive and tedious task of assessing quality in facilitating teacher peer production.

Our results suggest that indicators with little agreement among teachers should not be used as measures of quality. However, the two quality indicators with lots of teacher agreement showed moderate levels of agreement with trained OPERA’s performance. Therefore, it may be possible to use trained OPERA as part of the peer production process as an alternative to the expensive, multiple teacher rater approach.

In terms of limitations, we note that, overall, teacher ratings showed high means, with less discriminating power. This suggests that teachers may have needed additional training in applying these indicators. Alternatively, the six indicators and their definition may need refinement.

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